2018-Ph-H1-Q19

Section: Particles and Waves

Topic: Interference

Two coherent sound sources produce an interference pattern.
At a point of destructive interference, the path difference is:

A. 1 wavelength

B. ½ wavelength

C. 2 wavelengths

D. 0 wavelengths

Solution:

Destructive interference occurs when the path difference is **½ a wavelength**, or any odd multiple of it.

Answer: B

Revision Tip:

- **Constructive**: path difference = $n\lambda$
- **Destructive**: path difference = $(n+)\lambda$